

Royal Netherlands Institute for Sea Research

FAIR - UNECSS workshop

Best Practices for Making (Meta)Data FAIR

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Principes of Accessible

A1. (Meta)data are retrievable by their identifier using a standardised communications protocol

- A1.1 The protocol is open, free, and universally implementable
- A1.2 The protocol allows for an authentication and authorisation procedure, where necessary

A2. Metadata are accessible, even when the data are no longer available

What does this mean?

A1. (Meta)data are retrievable by their identifier using a standardised communications protocol

- A1.1 The protocol is open, free, and universally implementable
- A1.2 The protocol allows for an authentication and authorisation procedure, where necessary

In short:

(Meta)data needs to be possibly accessible via a Persistent Identifier to anyone with a **computer and an internet connection.**

Source: 'FAIR Principles: Interpretations and Implementation Considerations'. Jacobsen et al, Data Intelligence 2020; 2 (1-2): 10–29. doi: https://doi.org/10.1162/dint_r_00024

What about Restricted Data?

FAIR data ≠ OPEN data

Open data has to be freely available on the Internet, with no financial, legal or technical barriers.

For data to be **FAIR**, they only need to be **possibly accessible**, which does not mean that they have to be **open**.

For **restricted data** a data sharing agreement or data access committee can be used.

But you need to provide **clear conditions** under which data can be accessed.

What does this mean?

A2. Metadata are accessible, even when the data are no longer available

This means:

Metadata are **always openly accessible**, even if the data is restricted, under embargo or gone.

For data that may have been used and are referenced by others, it is important to have access to metadata that describes those resources.
This helps to understand their nature and their provenance, even when the relevant data are not available anymore.

Source: 'FAIR Principles: Interpretations and Implementation Considerations'. Jacobsen et al, Data Intelligence 2020; 2 (1-2): 10–29. doi: https://doi.org/10.1162/dint_r_00024

What about machine accessibility?

Machine accessibility is provided by data publishers and repositories

This can be done via an Application Programming Interface (API)

APIs are a way for different systems and applications to communicate with each other.
They can share data and functionality, and work together to perform tasks

Why is machine accessibility useful?

Machine accessibility helps with:

- Keeping data up to date
- Setting up better automation
- Keeps thing more repeatable
- Making data more findable

What can you do to make you data more accessible?

- Publish your data!
- Add a complete set of Metadata
- Make you data Interoperable.
- Make your data as **open as possible and as closed as necessary**

What to avoid?

Google Drive / Dropbox / OneDrive links

- Subject to commercial service terms
- “Link rot”, when data is moved/deleted, links break

"Data available from the corresponding author on reasonable request“

- email-as-protocol only when the contact route is explicit in the metadata and justified by sensitivity
- “Gabelica, M., Bojčić, R., Puljak, L. (2022). Many researchers were not compliant with their published data sharing statement. *Journal of Clinical Epidemiology* 150:33–41”